

BLUESTOCK FINANCE – CAPSTONE PROJECT

ABSTRACT

This project focuses on financial data analysis and portfolio optimization using mutual fund data from the Bluestock platform. The objective was to analyze fund performance, evaluate risk-return characteristics, and construct an optimal investment portfolio using the Markowitz Efficient Frontier framework. The study utilized historical Net Asset Value (NAV) data of selected funds, followed by preprocessing, return calculation, and covariance analysis. Portfolio optimization was performed by maximizing the Sharpe Ratio while incorporating a risk-free rate relevant to the Indian market.

The results demonstrate that optimal portfolio allocation significantly differs from naive diversification, emphasizing the importance of quantitative techniques in investment decision-making. Visualization tools such as Power BI were used to generate insights into fund performance trends, asset allocation, and risk exposure. The project highlights how data-driven financial modeling can improve portfolio efficiency and support informed investment strategies.

Index Terms— Portfolio Optimization, Markowitz Model, Mutual Funds, Risk-Return Analysis, Data Visualization

1. INTRODUCTION

In recent years, data-driven decision-making has become increasingly important in financial markets. Investors seek optimized portfolios that balance risk and return effectively. Mutual funds, being a popular investment instrument, provide diversified exposure but require careful analysis to maximize returns.

This project addresses the problem of selecting an optimal combination of mutual funds using quantitative techniques. Traditional investment strategies often rely on intuition or equal allocation, which may not yield optimal results. Portfolio optimization models such as the Markowitz Efficient Frontier provide a mathematical framework to achieve better outcomes.

In this project, historical mutual fund data was analyzed, and statistical methods were applied to derive optimal allocations. Additionally, visualization tools were used to present insights in an intuitive manner.

Report Structure: Section 2 defines the problem and objectives. Section 3 explains the methodology. Section 4 presents results. Section 5 discusses findings, and Section 6 concludes the work.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

Investors often struggle to allocate funds efficiently among multiple mutual funds due to uncertainty in returns and associated risks. The problem addressed in this project is:

How to construct an optimal portfolio of mutual funds that maximizes return for a given level of risk using historical data?

Key challenges include:

- Variability in fund performance
 - Correlation between assets
 - Risk estimation
 - Lack of quantitative allocation strategies
-

2.2 Objectives

1. To analyze historical NAV data of selected mutual funds
 2. To compute returns, volatility, and covariance
 3. To implement the Markowitz Efficient Frontier model
 4. To optimize portfolio allocation using Sharpe Ratio
 5. To visualize insights using Power BI dashboards
-

3. METHODOLOGY / APPROACH

3.1 Overall Workflow

The project follows this pipeline:

Data Collection → Data Cleaning → Return Calculation → Risk Analysis → Portfolio Optimization → Visualization

3.2 Technical Details

Data Processing

- Input: Historical NAV data of 5 mutual funds
- Frequency: Daily data
- Missing values handled using forward-fill

Return Calculation

Daily returns computed as:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Portfolio Optimization

The Markowitz model is defined as:

$$\text{Maximize Sharpe Ratio} = \frac{E[R_p] - R_f}{\sigma_p}$$

Where:

- R_p = Portfolio return
- R_f = 6% (India risk-free rate)
- σ_p = Portfolio volatility

Constraints:

- Weights sum to 1
- No short selling (weights ≥ 0)

Tools Used

- Python (NumPy, Pandas, SciPy)
- Power BI (Visualization)
- GitHub (Version Control)

4. EXPERIMENTS AND RESULTS

4.1 Experimental Setup

- Dataset: Mutual fund NAV data (2013 onwards)
- Assets: 5 funds (IDs: 119551–119555)
- Evaluation Metrics:
 - Mean Return
 - Volatility
 - Sharpe Ratio

4.2 Results

Optimal Portfolio Allocation

- Fund 119551: 5%
- Fund 119552: 5%
- Fund 119553: 5%
- Fund 119554: 50%
- Fund 119555: 35%

Key Observations

- Fund 119554 dominates due to high return-to-risk ratio
- Diversification improves stability
- Correlation between funds impacts allocation

5. DISCUSSION

The results indicate that optimized portfolios significantly outperform equal-weight strategies. The dominance of Fund 119554 suggests superior performance characteristics, while smaller allocations to other funds provide diversification benefits.

Trade-offs observed:

- Higher returns often come with increased risk
- Diversification reduces volatility but may limit returns

Limitations:

- Based on historical data only
 - Market conditions may change
 - Assumes constant risk-free rate
-

6. CONCLUSION AND FUTURE WORK

This project successfully demonstrates the application of portfolio optimization techniques to mutual fund data. The Markowitz Efficient Frontier provides a systematic approach to balancing risk and return.

Key Learnings

- Importance of covariance in portfolio design
- Role of risk-free rate in Sharpe optimization
- Value of visualization in financial analysis

Future Work

- Include more funds/assets
 - Apply machine learning models for prediction
 - Real-time portfolio optimization
 - Integration with live APIs
-

7. ARTIFACTS AND DEMONSTRATIONS

- GitHub Repository:
<https://github.com/nehasree1411/Bluestock-Data-Analysis-Capstone-Project>
- Project Outputs:
 - Portfolio Optimization Code
 - Visualization Reports (PNG)